



The University of Danang

**University of Science and Technology**

# MATHEMATICS FOR COMPUTER SCIENCE

## *Chap 1. Number theory*



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## References

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## Number theory

- **Primes and factors**
- Modular arithmetic
- Solving equations

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# Number theory

- **Number theory:**
  - is a branch of mathematics that studies integers.
  - Number theory is a fascinating field, because many questions involving integers are very difficult to solve even if they seem simple at first glance.

## Number theory

- As an example, consider the following equation:

$$x^3 + y^3 + z^3 = 33$$

- It is easy to find three real numbers  $x$ ,  $y$  and  $z$  that satisfy the equation. For example, we can choose  $x = 3, y = \sqrt[3]{3}, z = \sqrt[3]{3}$
- **However, it is an open problem in number theory if there are any three integers  $x$ ,  $y$  and  $z$  that would satisfy the equation**

## Primes and factors

- **Product** – An answer to a multiplication problem.
- **Factor** – a number that is multiplied by another to give a product.

$$2 \times 12 = 24$$

**Factors**

**Product**

## Primes and factors

- A number  $a$  is called a **factor** or a **divisor** of a number  $b$  if  $a$  divides  $b$ .
  - If  $a$  is a factor of  $b$ , we write  $a|b$ , and
  - *otherwise* we write  $a \nmid b$ .
  - For example, the factors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24.

## Primes and factors

$$3 \times 7 = 21$$

Factors: 3 and 7  
Product: 21

$$5 \times 6 = 30$$

Factors: 5 and 6  
Product: 30

## Primes and factors

- A number  $n > 1$  is a prime if its only **positive factors** are **1** and **n**.
  - 2, 3, 5, 7, 11, 13, 17 are **primes**
  - 35 is not a prime, because  $5 \times 7 = 35$
- **Composite number** – a number that has more than two factors.
  - 8 is Composite number: the factors of 8 are 1, 2, 4, 8.
- For every number  $n > 1$ , there is a **unique prime factorization**:

$$n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot p_3^{\alpha_3} \cdots p_k^{\alpha_k}$$

where  $p_1, p_2, \dots, p_k$  are distinct primes and  $\alpha_1, \alpha_2, \dots, \alpha_k$  are positive numbers

- For example, the prime factorization for 84 is

$$84 = 2^2 \cdot 3^1 \cdot 7^1$$

## Primes and factors

- The number of factors of a number  $n$  is

$$\tau(n) = \prod_{i=1}^k (\alpha_i + 1)$$

because for each prime  $p_i$ , there are  $(\alpha_i + 1)$  ways to choose how many times it appears in the factor.

- The prime factorization for 84 is  $84 = 2^2 \cdot 3^1 \cdot 7^1$
- The number of factors of 84 is  $\tau(84) = 3 \cdot 2 \cdot 2 = 12$ .
- The factors are 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42 and 84.

## Primes and factors

- The **sum of factors** of  $n$  is:

$$\sigma(n) = \prod_{i=1}^k (1 + p_i + \dots + p_i^{\alpha_i}) = \prod_{i=1}^k \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}$$

where the latter formula is based on **the geometric progression formula**.

- For example, the sum of factors of **84** is:

$$\sigma(84) = \frac{2^3-1}{2-1} \cdot \frac{3^2-1}{3-1} \cdot \frac{7^2-1}{7-1} = 7 \cdot 4 \cdot 8 = 224$$

## Primes and factors

- The **product of factors** of  $n$  is

$$\mu(n) = n^{\tau(n)/2}$$

because we can form  $\frac{\tau(n)}{2}$  pairs from the factors, each with product  $n$ .

- For example, the factors of 84 produce the pairs  $1 \cdot 84, 2 \cdot 42, 3 \cdot 28$ , etc., and
- The product of the factors is  $\mu(84) = 84^6 = 351298031616$

## Primes and factors

- A number  $n$  is called a **perfect number** if  $n = \sigma(n) - n$ , i.e.,
  - $n$  equals the sum of its factors between  $1$  and  $n - 1$ .
  - For example,  $28$  is a **perfect number**, because  $28 = 1 + 2 + 4 + 7 + 14$ .

## Number of primes

- It is easy to show that there is an infinite number of primes.
  - If the number of primes would be finite, we could construct a set  $P = \{p_1, p_2, \dots, p_n\}$  that would contain all the primes.
  - For example,  $p_1 = 2, p_2 = 3, p_3 = 5$ , and so on. However, using  $P$ , we could form a new prime

$$p_1 p_2 \cdots p_n + 1$$

that is larger than all elements in  $P$ . This is a contradiction, and the number of primes has to be infinite.

## Density of primes

- The density of primes means how often there are primes among the numbers.
  - Let  $\pi(n)$  denote the number of primes between 1 and  $n$ . For example,  $\pi(10) = 4$ , because there are 4 primes between 1 and 10 : 2, 3, 5 and 7.
  - It is possible to show that

$$\pi(n) \approx \frac{n}{\ln n}$$

which means that primes are quite frequent.

- For example, the number of primes between 1 and  $10^6$  is

$$\pi(10^6) = 78498 \text{ and } \frac{10^6}{\ln 10^6} \approx 72382$$

## Conjectures

- There are many conjectures involving primes. Most people think that the conjectures are true, but nobody has been able to prove them.
- For example, the following conjectures are famous:
  - **Goldbach's conjecture:** Each even integer  $n > 2$  can be represented as a sum  $n = a + b$  so that both  $a$  and  $b$  are primes.
  - **Twin prime conjecture:** There is an infinite number of pairs of the form  $\{p, p + 2\}$ , where both  $p$  and  $p + 2$  are primes.
  - **Legendre's conjecture:** There is always a prime between numbers  $n^2$  and  $(n + 1)^2$ , where  $n$  is any positive integer.

# Algorithms

- Basic algorithms
- Sieve of Eratosthenes
- Euclid's algorithm
- Euler's totient function



## Basic algorithms

- If a number  $n$  is not prime, it can be represented as a product  $a \cdot b$ , where  $a \leq \sqrt{n}$ , or  $b \leq \sqrt{n}$ , so it certainly has a factor between 2 and  $\lfloor \sqrt{n} \rfloor$ 
  - Using this observation, we can both test if a number is prime and find the prime factorization of a number in  $O(\sqrt{n})$  time.

*(The function  $\lfloor x \rfloor$  rounds the number  $x$  down to an integer, and the function  $\lceil x \rceil$  rounds the number  $x$  up to an integer:  $\lfloor \frac{3}{2} \rfloor = 1$  and  $\lceil \frac{3}{2} \rceil = 2$  )*

## Basic algorithms

- The following function *prime* checks if the given number  $n$  is prime. The function attempts to divide  $n$  by all numbers between 2 and  $\lfloor \sqrt{n} \rfloor$  and if none of them divides  $n$ , then  $n$  is prime.

```
bool prime(int n) {  
    if (n < 2) return false;  
    for (int x = 2; x*x <= n; x++) {  
        if (n%x == 0) return false;  
    }  
    return true;  
}
```

## Basic algorithms

- The following function *factors* constructs a vector that contains the prime factorization of  $n$ . The function divides  $n$  by its prime factors, and adds them to the vector. The process ends when the remaining number  $n$  has no factors between 2 and  $\lfloor \sqrt{n} \rfloor$ . If  $n > 1$ , it is prime and the last factor.

```
vector<int> factors(int n) {
    vector<int> f;
    for (int x = 2; x*x <= n; x++) {
        while (n%x == 0) {
            f.push_back(x);
            n /= x;
        }
    }
    if (n > 1) f.push_back(n);
    return f;
}
```

- Note that each prime factor appears in the vector as many times as it divides the number. For example,  $24 = 2^3 \cdot 3$ , so the result of the function is (2,2,2,3).

## Sieve of Eratosthenes

- The **sieve of Eratosthenes** is a preprocessing algorithm that builds an array using which we can efficiently check if a given number between  $2 \dots n$  is prime and, if it is not, find one prime factor of the number.
- The algorithm builds an array sieve whose positions  $2, 3, \dots, n$  are used. The value  $sieve[k] = 0$  means that  $k$  is prime, and the value  $sieve[k] \neq 0$  means that  $k$  is not a prime and one of its prime factors is  $sieve[k]$ .
- The algorithm iterates through the numbers  $2 \dots n$  one by one.
  - Always when a new prime  $x$  is found, the algorithm records that the multiples of  $x$  ( $2x, 3x, 4x, \dots$ ) are not primes

## Sieve of Eratosthenes

- For example, if  $n = 20$ , the array is as follows:

| 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 0 | 2 | 0 | 3 | 0 | 2 | 3 | 5  | 0  | 3  | 0  | 7  | 5  | 2  | 0  | 3  | 0  | 5  |

- The following code implements the sieve of Eratosthenes. The code assumes that each element of sieve is initially zero.

```
for (int x = 2; x <= n; x++) {
    if (sieve[x]) continue;
    for (int u = 2*x; u <= n; u += x) {
        sieve[u] = x;
    }
}
```

## Sieve of Eratosthenes

- The inner loop of the algorithm is executed  $n/x$  times for each value of  $x$ . Thus, an upper bound for the running time of the algorithm is the harmonic sum

$$\sum_{x=2}^n (n/x) = \frac{n}{2} + \frac{n}{3} + \frac{n}{4} + \dots + \frac{n}{n} = O(n \log n)$$

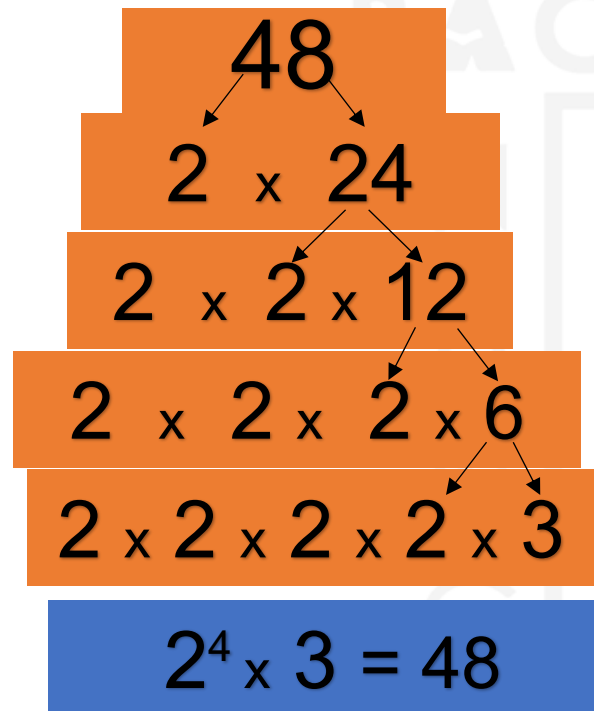
- In fact, the algorithm is more efficient, because the inner loop will be executed only if the number  $x$  is prime. It can be shown that the running time of the algorithm is only  $O(n \log(\log n))$ , a complexity very near to  $O(n)$ .

## How to do a Prime Factorization? Factor Tree Method

- **Step 1 – Write down any composite number.**
- **Step 2 – Start dividing by the prime #s (start with 2).**
  - If the composite number is divisible by 2, write it down and find the next factor.
  - If not, check if the factor is evenly divisible by 3, 5, 7, 9, etc.
- **Step 3 – Check the factors.**
  - If they are prime, you are done.
  - If they are not, proceed to Step 4.
- **Step 4 – Continue dividing.**
  - If one of the factors is divisible by 2, write it down and find the next factor.
  - If not, check if the factor is evenly divisible by 3, 5, 7, 9, etc.

# How to do a Prime Factorization? Factor Tree Method

- **Step 5 – Check the factors.**
  - If they are prime, proceed to Step 6.
  - If they are not, repeat Step 4.
- **Step 6 – Write the Prime Factorization in Exponential Form.**



## Euclid's algorithm

- The **greatest common divisor** of numbers  $a$  and  $b$ ,  $\gcd(a,b)$ , is the greatest number that divides both  $a$  and  $b$ , and
- The **least common multiple** of  $a$  and  $b$ ,  $\text{lcm}(a,b)$ , is the smallest number that is divisible by both  $a$  and  $b$ .
  - For example,  $\gcd(24,36) = 12$  and  $\text{lcm}(24,36) = 72$ .
- The greatest common divisor and the least common multiple are connected as follows:

$$\text{lcm}(a, b) = \frac{ab}{\gcd(a, b)}$$

## Euclid's algorithm

- Euclid's algorithm<sup>1</sup> provides an efficient way to find the greatest common divisor of two numbers. The algorithm is based on the following formula:

$$\gcd(a, b) = \begin{cases} a & b = 0 \\ \gcd(b, a \bmod b) & b \neq 0 \end{cases}$$

- For example:  $\gcd(24, 36) = \gcd(36, 24) = \gcd(24, 12) = \gcd(12, 0) = 12$ .

## Euclid's algorithm

- The algorithm can be implemented as follows:

```
int gcd(int a, int b) {  
    if (b == 0) return a;  
    return gcd(b, a%b);  
}
```

- It can be shown that Euclid's algorithm works in  $O(\log n)$  time, where  $n = \min(a, b)$ . The worst case for the algorithm is the case when  $a$  and  $b$  are consecutive Fibonacci numbers. For example,

$$\gcd(13,8) = \gcd(8,5) = \gcd(5,3) = \gcd(3,2) = \gcd(2,1) = \gcd(1,0) = 1$$

## Euler's totient function

- Numbers  $a$  and  $b$  are coprime if  $\gcd(a, b) = 1$ . Euler's totient function  $\varphi(n)$  gives the number of coprime numbers to  $n$  between 1 and  $n$ .
  - For example,  $\varphi(12) = 4$ , because 1, 5, 7 and 11 are coprime to 12.
  - The value of  $\varphi(n)$  can be calculated from the prime factorization of  $n$  using the formula

$$\varphi(n) = \prod_{i=1}^k p_i^{\alpha_i-1} (p_i - 1)$$

- For example,  $\varphi(12) = 2^1 \cdot (2 - 1) \cdot 3^0 \cdot (3 - 1) = 4$ .
- Note that  $\varphi(n) = n - 1$  if  $n$  is prime.

## Number theory

- Primes and factors
- **Modular arithmetic**
- Solving equations

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## Modular arithmetic

- In modular arithmetic, the set of numbers is limited so that only numbers  $0, 1, 2, \dots, m - 1$  are used, where  $m$  is a constant. Each number  $x$  is represented by the number  $x \bmod m$ :
  - The remainder after dividing  $x$  by  $m$ .
  - For example, if  $m = 17$ , then 75 is represented by  $75 \bmod 17 = 7$ .

## Modular arithmetic

- Often we can take remainders before doing calculations. In particular, the following formulas hold:

$$(x + y) \bmod m = (x \bmod m + y \bmod m) \bmod m$$

$$(x - y) \bmod m = (x \bmod m - y \bmod m) \bmod m$$

$$(x \cdot y) \bmod m = (x \bmod m \cdot y \bmod m) \bmod m$$

$$x^n \bmod m = (x \bmod m)^n \bmod m$$

## Modular exponentiation

- There is often need to efficiently calculate the value of  $x^n \bmod m$ . This can be done in  $O(\log n)$  time using the following recursion:

$$x^n = \begin{cases} 1 & n = 0 \\ x^{n/2} \cdot x^{n/2} & n \text{ is even} \\ x^{n-1} \cdot x & n \text{ is odd} \end{cases}$$

- It is important that in the case of an even  $n$ , the value of  $x^{n/2}$  is calculated only once. This guarantees that the time complexity of the algorithm is  $O(\log n)$ , because  $n$  is always halved when it is even.

## Modular exponentiation

- The following function calculates the value of  $x^n \bmod m$ :

```
int modpow(int x, int n, int m) {  
    if (n == 0) return 1%m;  
    long long u = modpow(x, n/2, m);  
    u = (u*u)%m;  
    if (n%2 == 1) u = (u*x)%m;  
    return u;  
}
```

## Fermat's theorem and Euler's theorem

- **Fermat's theorem** states that:

$$x^{m-1} \bmod m = 1$$

when  $m$  is prime and  $x$  and  $m$  are coprime. This also yields

$$x^k \bmod m = x^{k \bmod (m-1)} \bmod m$$

- **Euler's theorem** states that

$$x^{\varphi(m)} \bmod m = 1$$

when  $x$  and  $m$  are coprime. **Fermat's theorem** follows from Euler's theorem, because if  $m$  is a prime, then  $\varphi(m) = m - 1$ .

## Modular inverse

- The inverse of  $x$  modulo  $m$  is a number  $x^{-1}$  such that

$$xx^{-1} \bmod m = 1$$

- For example, if  $x = 6$  and  $m = 17$ , then  $x^{-1} = 3$ , because  $6 \cdot 3 \bmod 17 = 1$ .
- Using modular inverses, we can divide numbers modulo  $m$ , because division by  $x$  corresponds to multiplication by  $x^{-1}$ .
  - For example, to evaluate the value of  $36/6 \bmod 17$ , we can use the formula  $2 \cdot 3 \bmod 17$ , because  $36 \bmod 17 = 2$  and  $6^{-1} \bmod 17 = 3$ .
  - However, a modular inverse does not always exist. For example, if  $x = 2$  and  $m = 4$ , the equation  $xx^{-1} \bmod m = 1$  cannot be solved, because all multiples of 2 are even and the remainder can never be 1 when  $m = 4$ . **It turns out that the value of  $x^{-1} \bmod m$  can be calculated exactly when  $x$  and  $m$  are coprime.**

## Modular inverse

- If a modular inverse exists, it can be calculated using the formula

$$x^{-1} = x^{\varphi(m)-1}$$

- If  $m$  is prime, the formula becomes  $x^{-1} = x^{m-2}$
- For example,  $6^{-1} \bmod 17 = 6^{17-2} \bmod 17 = 3$ .
- **This formula allows us to efficiently calculate modular inverses using the modular exponentiation algorithm.**

## Modular inverse

- The formula can be derived using **Euler's theorem**.
  - First, the modular inverse should satisfy the following equation:

$$xx^{-1} \bmod m = 1$$

- On the other hand, according to Euler's theorem,

$$x^{\varphi(m)} \bmod m = xx^{\varphi(m)-1} \bmod m = 1$$

so the numbers  $x^{-1}$  and  $x^{\varphi(m)-1}$  are equal

## Computer arithmetic

- In programming, unsigned integers are represented modulo  $2^k$ , where  $k$  is the number of bits of the data type.
  - A usual consequence of this is that a number wraps around if it becomes too large.
  - For example, in C++, numbers of type unsigned int are represented modulo  $2^{32}$ .
  - The following code declares an **unsigned int variable** whose value is 123456789. After this, the value will be multiplied by itself, and the result is  $123456789^2 \bmod 2^{32} = 2537071545$ .

```
unsigned int x = 123456789;  
cout << x*x << "\n"; // 2537071545
```

## Number theory

- Primes and factors
- Modular arithmetic
- **Solving equations**



## Solving equations- Diophantine equations

- **Diophantine equations**

- A **Diophantine equation** is an equation of the form

$$ax + by = c$$

where  $a$ ,  $b$  and  $c$  are constants and the values of  $x$  and  $y$  should be found. Each number in the equation has to be an integer.

- For example, one solution for the equation  $5x + 2y = 11$  is  $x = 3$  and  $y = -2$ .
- We can efficiently solve a Diophantine equation by using Euclid's algorithm.
  - It turns out that we can extend Euclid's algorithm so that it will find numbers  $x$  and  $y$  that satisfy the following equation:  $ax + by = \gcd(a, b)$
  - A **Diophantine equation** can be solved if  $c$  is divisible by  $\gcd(a, b)$ , and otherwise it cannot be solved.

## Solving equations- Diophantine equations

- As an example, let us find numbers  $x$  and  $y$  that satisfy the following equation:

$$39x + 15y = 12$$

- The equation can be solved, because  $\gcd(39,15) = 3$  and  $3|12$ .
- When Euclid's algorithm calculates the greatest common divisor of 39 and 15, it produces the following sequence of function calls:

$$\gcd(39,15) = \gcd(15,9) = \gcd(9,6) = \gcd(6,3) = \gcd(3,0) = 3$$

- This corresponds to the following equations:

$$39 - 2 \cdot 15 = 9; 15 - 1 \cdot 9 = 6; 9 - 1 \cdot 6 = 3$$

- Using these equations, we can derive:  $39 \cdot 2 - 15 \cdot (-5) = 3$  and by multiplying this by 4, the result is  $39 \cdot 8 + 15 \cdot (-20) = 12$ , so a solution to the equation is  $x = 8$  and  $y = -20$ .

## Solving equations- Diophantine equations

- A solution to a Diophantine equation is not unique, because we can form an infinite number of solutions if we know one solution. If a pair  $(x, y)$  is a solution, then also all pairs:

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right)$$

*are solutions, where  $k$  is any integer*

## Solving equations-Chinese remainder theorem

- The **Chinese remainder theorem** solves a group of equations of the form

$$x = a_1 \bmod m_1$$

$$x = a_2 \bmod m_2$$

...

$$x = a_n \bmod m_n$$

where all pairs of  $m_1, m_2, \dots, m_n$  are coprime.

- Let  $x_m^{-1}$  be the inverse of  $x \bmod m$ , and  $X_k = \frac{m_1 m_2 \dots m_k}{m_k}$
- Using this notation, a solution to the equations is

$$x = a_1 X_1 X_{1m_1}^{-1} + a_2 X_2 X_{2m_2}^{-1} + \dots + a_n X_n X_{nm_n}^{-1}$$

## Solving equations-Chinese remainder theorem

- In this solution, for each  $k = 1, 2, \dots, n$ ,

$$a_k X_k X_{k m_k}^{-1} \bmod m_k = a_k$$

because

$$X_k X_{k m_k}^{-1} \bmod m_k = 1$$

- Since all other terms in the sum are divisible by  $m_k$ , they have no effect on the remainder, and  $x \bmod m_k = a_k$ .
  - For example, a solution for

$$x = 3 \bmod 5$$

$$x = 4 \bmod 7$$

$$x = 2 \bmod 3$$

is  $3 \cdot 21 \cdot 1 + 4 \cdot 15 \cdot 1 + 2 \cdot 35 \cdot 2 = 263$ . Once we have found a solution  $x$ , we can create an infinite number of other solutions, because all numbers of the form  $x + m_1 m_2 \dots m_n$  are solutions

## Solving equations-Other results

- **Lagrange's theorem**

- Lagrange's theorem states that every positive integer can be represented as a sum of four squares, i.e.,  $a^2 + b^2 + c^2 + d^2$ . For example, the number 123 can be represented as the sum  $8^2 + 5^2 + 5^2 + 3^2$ .

- **Zeckendorf's theorem**

- Zeckendorf's theorem states that every positive integer has a unique representation as a sum of Fibonacci numbers such that no two numbers are equal or consecutive Fibonacci numbers.
- For example, the number 74 can be represented as the sum  $55 + 13 + 5 + 1$ .

## Solving equations-Other results

- **Pythagorean triples**

- A Pythagorean triple is a triple  $(a, b, c)$  that satisfies the Pythagorean theorem  $a^2 + b^2 = c^2$ , which means that there is a right triangle with side lengths  $a$ ,  $b$  and  $c$ . For example,  $(3, 4, 5)$  is a Pythagorean triple.
- If  $(a, b, c)$  is a Pythagorean triple, all triples of the form  $(ka, kb, kc)$  are also Pythagorean triples where  $k > 1$ .
- A Pythagorean triple is primitive if  $a$ ,  $b$  and  $c$  are coprime, and all Pythagorean triples can be constructed from primitive triples using a multiplier  $k$ .

## Solving equations-Other results

- **Pythagorean triples**

- **Euclid's formula** can be used to produce all primitive Pythagorean triples.  
Each such triple is of the form

$$(n^2 - m^2, 2nm, n^2 + m^2),$$

where  $0 < m < n$ ,  $n$  and  $m$  are coprime and at least one of  $n$  and  $m$  is even. For example, when  $m = 1$  and  $n = 2$ , the formula produces the smallest Pythagorean triple:

$$(2^2 - 1^2, 2 \cdot 2 \cdot 1, 2^2 + 1^2) = (3, 4, 5)$$

## Solving equations-Other results

- **Wilson's theorem**

- Wilson's theorem states that a number  $n$  is prime exactly when

$$(n - 1)! \bmod n = n - 1$$

- For example, the number 11 is prime, because

$$10! \bmod 11 = 10$$

- and the number 12 is not prime, because

$$11! \bmod 12 = 0 \neq 11$$

- Wilson's theorem can be used to find out whether a number is prime.
- However, in practice, the theorem cannot be applied to large values of  $n$ , because it is difficult to calculate values of  $(n - 1)!$  when  $n$  is large.